

Brenda Murillo

Palmdale, CA

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EDUCATION

California State University, Bakersfield

Bakersfield, CA

Bachelor of Science, Computer Science | GPA: 3.65/4.0

Expected May 2027

Leadership: Vice President, Robotics Club | MESA Mentor

Relevant Coursework: Artificial Intelligence, Linear Algebra, Database Systems, Data Structures, Algorithm Analysis

TECHNICAL SKILLS

Programming: Python, SQL, R, TypeScript

Machine Learning & Data Science: scikit-learn, XGBoost, pandas, NumPy, feature engineering, model tuning, model evaluation, statistical analysis

AI: Computer Vision, LLMs, RAG, Agentic AI, LangGraph, Chroma, Multimodal AI

Databases & Development: PostgreSQL, MySQL, MongoDB, Supabase, React Native, Git/GitHub, Gradio

Visualization: Tableau, Matplotlib, Excel, ggplot2

EXPERIENCE

Let's Talk | Industry-Sponsored Senior Capstone

Remote

AI/Software Developer

August 2026 – Present

- Developing a secure co-parenting communication platform using React Native, TypeScript, Supabase, and PostgreSQL, with real-time messaging, authentication, and family-based account linking.
- Implementing database-level authorization with PostgreSQL Row Level Security (RLS) and immutable message history to protect sensitive communications and support future court-aligned records.
- Contributing to AI-powered de-escalation and legal Q&A features using LLM/RAG workflows, including neutral message rewording and source-grounded responses over curated legal information.

Handshake AI

Remote

AI Data Evaluator

February 2026 – Present

- Evaluated text, image, and audio datasets to improve machine learning training quality.
- Validated annotations and identified inconsistencies to support high-quality AI model development.

Data Analyst Workshop

Lancaster, CA

Data Analyst Trainee

June 2026

- Built 10+ interactive Tableau and Excel dashboards; used SQL and Python to clean, validate, and analyze datasets for trends and decision-making.
- Applied EDA, preprocessing, and statistical analysis across 20+ analytical projects, improving data quality and communicating findings through visualizations.

RESEARCH & MACHINE LEARNING PROJECTS

Adaptive and Explainable Ball Tracking | IEEE MIUCC 2026 | Co-author

- Co-authored IEEE-published computer-vision research using CLAHE, LAB segmentation, contour validation, motion stability, and confidence-based decision-making to improve humanoid robot tracking under varying lighting and distance conditions.

Predicting Employee Turnover Using Machine Learning | Research Project | Manuscript in Preparation

- Developed and compared Logistic Regression, Random Forest, and XGBoost models using a dataset of 15,000+ employee records; performed preprocessing, feature engineering, and hyperparameter tuning.
- Evaluated model performance using ROC-AUC, precision, recall, F1-score, and confusion matrices.

Ripeness Bench - Computer Vision & Agentic AI | Collaborative Project

- Built an agentic computer-vision system using specialized detection, ripeness, and recommendation agents to analyze two fruits from a single image and generate individualized recommendations.
- Implemented both offline rule-based vision and AI vision modes, integrated Raspberry Pi image capture and a web interface, and logged scan results for evaluation and future analysis.

PrepPal - AI Academic Assistant | Independent Project

- Developed an agentic AI academic assistant with web and arXiv search, calculation, GPA analysis, and RAG-based retrieval over student documents to provide personalized academic support.
- Implemented conversational and semantic memory using LangGraph and Chroma, with embedding-based retrieval, tool-input guardrails, tool-call logging, and an interactive Gradio interface.